

Structural Understanding as an Invariance Condition

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Executive Summary

This paper addresses a foundational question in the ontology of systems: under what conditions may a property be legitimately attributed to a system itself, rather than merely to its observable outputs? The analysis proceeds from a strict formal distinction between behavioural manifestation and structural instantiation. Let S denote a system, O_t an observation under interaction t , and F a candidate property. The central premise is the non-identity:

$$F(S) \neq F(O_t).$$

Consistent or even universal observation of $F(O_t)$ does not entail that F is a structural property of S . Performance adequacy and ontological classification operate under different logical regimes. The former quantifies over interactions; the latter quantifies over admissible system states.

To prevent conflation of these domains, the paper introduces a binary admissibility classification $C_s(S) \in \{0,1\}$. Structural attribution requires satisfaction of necessary predicates and is governed by necessary-condition logic. The absence of a required structural condition is sufficient to deny admissibility. Structural classification is therefore categorical rather than scalar.

The key theoretical contribution is the formalisation of an invariance condition for structural properties. A reset operator $T: State(S) \rightarrow s_0$, defined as an identity-preserving state transformation, is introduced to test whether a candidate property survives normalisation of internal configuration. If there exists a state s such that:

$$F(s) = 1 \text{ and } F(T(s)) = 0,$$

then F is state-dependent and cannot be treated as structural. Structural properties must satisfy:

$$F \in Struct(S) \iff \forall T \in \mathcal{T}(S), \forall s \in State(S), F(T(s)) = F(s)$$

Understanding, when construed as a system-level predicate, therefore becomes an invariance condition. Correct responses, reproducibility, and task performance are insufficient grounds for ontological attribution.

An empirical probe using inference systems is included solely to illustrate this admissibility logic under controlled transformation scenarios. The systems consistently deny structural classification when invariance is violated. No claims are made regarding comparative capability, architecture, or internal mechanism. The systems function as instruments revealing the structure of property attribution under reset conditions.

The paper's contribution is conceptually bounded. It does not determine whether any specific system possesses structural understanding. It clarifies the logical condition that must be satisfied if such a claim is to be admissible. Structural understanding, when construed as a structural predicate, is defined by invariance under identity-preserving state transformation.

Abstract

When may a property be legitimately attributed to a system rather than to its observable behaviour? This paper addresses that question by formalising the distinction between performance-based evaluation and structural classification. Let S denote a system, O_t an observation under interaction t , and F a candidate property. The central ontological problem arises from the non-identity $F(S) \neq F(O_t)$. Consistent manifestation of F across observed interactions does not, by itself, entail structural instantiation.

To resolve this ambiguity, the paper introduces a binary admissibility classification $C_s(S) \in \{0,1\}$, governed by necessary-condition logic. Structural attribution requires satisfaction of invariant relational predicates, while the failure of any necessary condition suffices to deny admissibility. A reset operator $T: State(S) \rightarrow s_0$, defined as an identity-preserving state transformation, is employed to test structural stability. If there exists a state s such that $F(s) = 1$ and $F(T(s)) = 0$, the property is state-dependent and therefore not structural.

From this analysis follows the central thesis: structural understanding, if treated as a system-level predicate, must satisfy an invariance condition under admissible state transformations. Behavioural regularity, reproducibility, or task success are insufficient grounds for ontological attribution. Structural properties are defined by stability across transformation, not by accumulation of correct responses.

The contribution of the paper is not an evaluation of system performance, but a formal clarification of the admissibility conditions under which structural predicates may be ascribed. Understanding, when construed structurally, is an invariance condition.

Scope and Synopsis

This paper develops a formal account of the conditions under which a property may be legitimately attributed to a system rather than to its observable outputs. Its scope is strictly ontological. It does not evaluate system performance, compare architectures, speculate about internal mechanisms, or engage broader debates concerning intelligence, consciousness, or general capability. The analysis is confined to the logical structure of property attribution.

The central problem addressed is the distinction between behavioural manifestation and structural instantiation. Let S denote a system, O_t an observation under interaction t , and F a candidate property. The paper begins from the formal non-identity $F(S) \neq F(O_t)$, and examines the inferential conditions under which one may pass from consistent observation of $F(O_t)$ to admissible attribution of $F(S)$.

A binary admissibility classification $C_s(S) \in \{0,1\}$ is introduced to prevent ontological inflation based on performance evidence alone. Structural attribution is governed by necessary-condition logic: positive classification requires satisfaction of invariant relational predicates, while the absence of any necessary condition suffices for denial. The analysis does not specify empirical measurement procedures for these predicates; it establishes only their logical role within structural classification.

To bridge the gap between interaction-level observation and system-level ontology, the paper introduces a reset operator $T: State(S) \rightarrow s_0$, defined as an identity-preserving state transformation. The reset serves as a structural probe. If there exists a state s such that $F(s) = 1$ and $F(T(s)) = 0$,

then F is state-dependent rather than constitutive. Structural properties must therefore satisfy an invariance condition:

$$F \in Struct(S) \Leftrightarrow \forall T \in \mathcal{T}(S), \forall s \in State(S), F(T(s)) = F(s)$$

The empirical component of the paper is strictly instrumental. A set of inference systems is used not to compare performance or rank capability, but to observe how property attribution behaves under controlled transformation scenarios. The systems consistently deny structural classification when invariance is violated, thereby illustrating the admissibility logic formalised in the theoretical sections.

The paper's contribution is limited and precise. It does not claim to determine whether any particular system possesses structural understanding. It clarifies the logical condition that must be satisfied if such a claim is to be admissible. Structural understanding, when construed as a system-level predicate, is defined by invariance under identity-preserving state transformation.

1. The Ontological Problem of Property Attribution

A persistent difficulty in the philosophy of science concerns the conditions under which a property may legitimately be attributed to a system rather than to its observable behaviour. This distinction, though often acknowledged in principle, is routinely collapsed in practice. The present analysis begins by formalising that distinction.

Let S denote a system.

Let O_t denote an observation of S under interaction t .

Let F denote a property.

We immediately distinguish between the attribution of a property to a system and the attribution of a property to an observation:

$$F(S) \neq F(O_t)$$

The expression $F(O_t)$ asserts that a given observation exhibits property F . The expression $F(S)$ asserts that the system itself instantiates F as a structural feature. These are logically distinct claims. The former concerns behavioural manifestation under specific interaction conditions; the latter concerns constitutive structure.

This distinction is not merely terminological. It marks the boundary between performance-based assessment and ontological attribution.

In empirical inquiry, properties are typically accessed through observation. One observes outputs, measurements, behaviours, or responses, and from these infers underlying characteristics. The inferential move from repeated observation to structural attribution is, however, not logically trivial. It presupposes that behavioural regularity suffices to establish structural instantiation. Whether such sufficiency holds is precisely the issue at stake.

The logical form of the problematic inference may be expressed as follows:

$$(\forall t, F(O_t)) \Rightarrow F(S)$$

That is, from the fact that every observed interaction exhibits property F , one infers that the system possesses F as a structural property. This inference is not deductively valid. At most, it establishes behavioural consistency under the observed range of interactions. It does not, by itself, establish invariance across all relevant system states.

The difficulty becomes especially acute when the property in question purports to be structural rather than dispositional. A structural property is not merely a tendency to produce certain outputs under certain inputs. It is a feature of the system's internal organisation that remains stable under permissible transformations of state.

Accordingly, we introduce a preliminary distinction between two logical types of property:

1. A performance property, defined through existence of observed manifestation:

$$\exists t: F(O_t)$$

2. A structural property, defined through attribution to the system as such:

$$F(S)$$

A performance property requires only that there exists at least one interaction under which the property is manifested. A structural property, by contrast, requires that the property be constitutive of the system's organisation, not contingent upon particular transient configurations.

The historical tendency has been to treat certain complex properties as performance properties while speaking of them as though they were structural. This conflation generates ontological ambiguity. A system may reliably produce outputs that resemble the manifestation of a given property without that property being structurally instantiated.

To avoid this ambiguity, one must specify the admissibility conditions under which a behavioural regularity licenses structural attribution. Without such conditions, the attribution of structural properties risks being reducible to empirical generalisation over finite observations.

The present paper advances a precise thesis:

Structural properties require invariance under admissible transformations of system state. Behavioural regularity alone is insufficient to establish structural instantiation.

This thesis will be formalised in subsequent sections through the introduction of an invariance condition and a reset-based transformation operator. For the moment, it suffices to establish the ontological problem in its minimal form: the non-identity of property attribution to observations and property attribution to systems.

The remainder of the analysis proceeds from this distinction.

2. Classification versus Performance

The distinction introduced in the preceding section allows for a more precise articulation of two fundamentally different modes of evaluation: performance assessment and ontological

classification. Though often treated as interchangeable in empirical practice, they operate under distinct logical regimes.

A performance assessment concerns the observable behaviour of a system under specified interaction conditions. Formally, it is satisfied whenever there exists at least one interaction t such that the observed output exhibits property F :

$$\exists t: F(O_t)$$

In its stronger form, performance assessment may involve a range of interactions:

$$\forall t \in T_{obs}: F(O_t)$$

where T_{obs} denotes the set of observed interaction contexts. Even in this stronger case, the quantification remains restricted to the domain of observation. The conclusion pertains exclusively to behavioural regularity within that domain.

Ontological classification, by contrast, does not concern the frequency or consistency of observed manifestations. It concerns whether the property F is admissibly attributable to the system as such. The relevant claim is:

$$F(S)$$

This is not an empirical summary of observed responses. It is a categorical assignment. To assert $F(S)$ is to classify the system as belonging to the extension of systems that structurally instantiate F .

The logical gap between these two modes of evaluation becomes evident when one considers the following inference schema:

$$(\forall t \in T_{obs}, F(O_t)) \not\Rightarrow F(S)$$

Even perfect behavioural consistency across all observed contexts does not entail structural instantiation. The inference fails because behavioural consistency is compatible with multiple underlying generative structures. A system may produce outputs satisfying F due to contingent configuration, temporary state alignment, external scaffolding, or statistical correlation without possessing the internal relational organisation required for structural attribution.

The distinction may be clarified by introducing the notion of admissibility. A structural classification requires that the property in question satisfy criteria that render it admissible as a system-level predicate. Performance evidence may be necessary for such admissibility, but it is not sufficient.

This difference can be expressed through a separation of quantificational domains:

- Performance claims quantify over interactions.
- Structural claims quantify over system states.

Performance logic operates over the space of observable input–output episodes. Structural logic operates over the space of admissible system configurations.

If $State(S)$ denotes the set of possible states of system S , then structural attribution implicitly concerns the behaviour of F across that state space, not merely across a subset of observed interactions. Performance assessment examines:

$$F(O_t) \text{ for } t \in T_{obs}$$

Structural classification requires consideration of:

$$F(s) \text{ for } s \in State(S)$$

The two are not coextensive. Observational completeness does not imply state-space coverage.

Historically, properties such as intelligence, competence, or understanding have frequently been evaluated under performance criteria while being described in structural terms. This practice introduces a modal inconsistency: a property is treated as though it were invariant across states, yet is established solely through episodic observation.

The present analysis rejects that conflation. A structural property must be defined in terms of conditions that hold independently of particular observational episodes. It must be stable under transformations that preserve the identity of the system while altering its transient configuration.

This requirement motivates the introduction of an invariance condition. Before formalising that condition, however, it is necessary to specify the minimal admissibility framework within which structural classification may be meaningfully articulated. That framework is introduced in the following section.

3. A Minimal Admissibility Framework

To formalise the distinction between performance evidence and structural attribution, we introduce a minimal admissibility classification. The purpose of this framework is not to measure degree or capacity, but to determine whether a given structural predicate may be attributed to a system at all.

Let S denote a system.

Define a binary structural classification function:

$$C_s(S) \in \{0,1\}$$

The value $C_s(S) = 1$ indicates that the system satisfies the admissibility conditions for structural attribution of the relevant property. The value $C_s(S) = 0$ indicates that such attribution is not warranted.

This function is not scalar. It does not admit gradation. It does not represent magnitude, efficiency, or performance quality. It represents a categorical classification.

We further introduce two necessary structural conditions, denoted $\Phi_i(S)$ and $R_g(S)$, without yet specifying operational measurement procedures. These are treated here solely as logical predicates over systems.

The admissibility condition is defined as:

$$C_s(S) = 1 \Rightarrow (\Phi_i(S) \wedge R_g(S))$$

That is, if structural classification is positive, then both necessary structural predicates must hold.

The contrapositive form is of central importance:

$$\neg R_g(S) \Rightarrow C_s(S) = 0$$

The absence of a necessary structural condition is sufficient to deny admissibility. Structural attribution is therefore governed by a necessary-condition logic rather than a performance-accumulation logic.

Two features of this framework must be emphasised.

First, the classification function C_s is not an empirical score. It is a logical admissibility operator. Its purpose is to prevent unwarranted ontological inflation based on behavioural evidence alone.

Second, the framework is asymmetrical. Positive classification requires satisfaction of necessary structural conditions. Negative classification requires only the demonstration that at least one necessary condition fails. In this respect, structural attribution operates under a principle of structural conservatism: absence of required structure is decisive.

At this stage, no assumptions are made regarding how Φ_i or R_g are empirically assessed. The present section establishes only their logical role as necessary predicates within a binary admissibility regime.

The relevance of this framework will become clear once the observational problem is reintroduced under formal interaction mapping. The next section therefore turns to the structure of interaction and the limits of observational inference.

4. The Observational Problem

We now return to the distinction between system and observation, and formalise the structure of interaction through which properties are typically inferred.

Let interaction at time t be represented as a mapping:

$$\sigma_t: S \rightarrow O_t$$

where S denotes the system and O_t denotes the observable output produced under interaction t . The mapping σ_t does not expose the internal constitution of S ; it yields only its behavioural manifestation under specific conditions.

Suppose that a property F is consistently observed across all interactions within an observational domain T_{obs} :

$$\forall t \in T_{obs}: F(O_t)$$

The inferential temptation is to conclude:

$$F(S)$$

However, this inference is not deductively valid. Formally,

$$(\forall t \in T_{obs}, F(O_t)) \not\Rightarrow F(S)$$

The reason is structural rather than epistemic. The domain of quantification in the antecedent concerns interactions; the domain of the consequent concerns system constitution. These domains are not logically equivalent.

Observational universality over T_{obs} guarantees only behavioural regularity within the sampled interaction space. It does not establish invariance across the full space of admissible system states. In particular, it leaves open at least three possibilities:

1. The property F may arise from transient state configurations rather than from constitutive organisation.
2. The property F may depend on interaction scaffolding external to the system.
3. The property F may fail under transformations that preserve system identity while altering internal state.

To clarify the third point, let $State(S)$ denote the set of possible internal states of the system. Observational consistency over interactions does not entail stability over state transitions. A system may produce outputs satisfying F across all observed interactions while still lacking structural invariance across its own admissible configurations.

The observational problem therefore consists in the following structural asymmetry:

Observation quantifies over interactions.

Structural attribution quantifies over system states.

Without a principled bridge between these domains, performance evidence cannot warrant ontological classification.

The inadequacy of observational sufficiency becomes especially clear when one considers counterfactual state transformations. If there exists a state $s \in State(S)$ such that the system fails to exhibit F , even while remaining the same system, then F cannot be constitutive of its structure.

Thus, structural attribution requires more than behavioural persistence across observed episodes. It requires stability under transformations that preserve system identity. The formal introduction of such transformations provides the necessary bridge between behavioural observation and ontological classification.

This motivates the introduction of a reset operator in the following section.

5. The Reset Operator

The observational problem identified in the previous section demonstrates that behavioural regularity over interactions does not suffice for structural attribution. To evaluate structural

admissibility, one must introduce a transformation that operates not at the level of interaction, but at the level of system state.

Let $State(S)$ denote the admissible internal states of system S .

Define a transformation:

$$T: State(S) \rightarrow s_0$$

where s_0 denotes a designated baseline state. The transformation T preserves system identity while altering internal configuration. That is, the system remains the same system S , but its transient state is normalised.

The reset operator does not destroy the system. It does not replace it. It does not introduce external scaffolding. It alters only internal state variables while preserving structural identity.

Now suppose that there exists a state $s \in State(S)$ such that:

$$F(s) = 1$$

That is, the property F is manifested in state s .

After application of the reset operator, consider the property evaluated at the transformed state:

$$F(T(s))$$

If the following holds:

$$F(s) = 1 \text{ and } F(T(s)) = 0$$

then we may formally state:

$$\exists s: F(s) \wedge \neg F(T(s))$$

This existential statement has direct ontological consequences. It establishes that the property F is state-dependent rather than structurally invariant. The system can exhibit F in at least one configuration while failing to exhibit F under a permissible transformation that preserves system identity.

Therefore, F cannot be constitutive of the system's structure.

The reset operator thus functions as a structural probe. It introduces a controlled state transformation that tests whether a candidate property survives identity-preserving normalisation.

Importantly, the reset test does not require exhaustive exploration of all states. It requires only the demonstration of at least one counterexample under admissible transformation. Because structural attribution is governed by necessary-condition logic, the existence of a single violating state suffices to deny admissibility.

This asymmetry mirrors the contrapositive structure introduced in Section 3:

$$\neg R_g(S) \Rightarrow C_s(S) = 0$$

Structural classification is falsifiable through state transformation. It is not confirmable through behavioural accumulation alone.

The reset operator therefore provides the missing bridge between observation and ontology. It converts the abstract requirement of state-space stability into a formally testable condition.

The next section makes explicit the principle that the reset operator implicitly encodes: structural properties must be invariant under admissible state transformations.

6. The Invariance Principle

The reset analysis introduced in the previous section reveals a necessary structural condition for admissible property attribution. That condition may now be stated explicitly.

A property qualifies as structural if and only if it is invariant under admissible transformations of system state that preserve system identity.

Formally, let $T \in \mathcal{T}(S)$, where $\mathcal{T}(S)$ denotes the class of admissible identity-preserving state transformations of S . Then:

$$F \in Struct(S) \Leftrightarrow \forall T \in \mathcal{T}(S), \forall s \in State(S), F(T(s)) = F(s)$$

for all admissible states $s \in State(S)$.

This condition defines structurality in terms of invariance. A property that varies under permissible state transformation cannot be constitutive of the system's organisation. It may be episodic, dispositional, or context-dependent, but it is not structural.

The reset operator introduced earlier is a particular instance of such a transformation. It serves as a diagnostic case:

If there exists at least one admissible transformation T and at least one state s such that:

$$F(T(s)) \neq F(s),$$

then it follows that:

$$F \notin Struct(S).$$

The inference is immediate. Structural properties must survive normalisation. If they do not, they fail the invariance condition.

This formulation establishes several important consequences.

First, structural attribution becomes transformation-relative rather than observation-relative. The relevant domain of quantification is no longer the set of observed interactions, but the set of admissible state transformations.

Second, structural classification acquires a falsifiability condition. Demonstrating non-invariance under a single admissible transformation suffices to deny structural status. Confirmation, by contrast, requires invariance across the relevant transformation class.

Third, the principle explains why behavioural regularity is insufficient. Observational persistence across interactions may reflect stability within a restricted subset of states, while failing under transformation. Invariance, not frequency, is the decisive criterion.

The invariance condition therefore resolves the ontological ambiguity identified in Section 1. It provides a principled bridge between performance evidence and structural attribution.

The central thesis of this paper may now be stated without qualification:

Structural understanding, if it is to be treated as a system-level property, must satisfy an invariance condition.

Understanding is not defined by the existence of correct responses. It is defined by stability of relational organisation across admissible transformations of state.

The remainder of the paper examines how this invariance condition may be probed in practice and what follows from its application.

6.1 Formal Definitions and Structural Conditions

To eliminate residual ambiguity, we now introduce explicit definitions of the state space, admissible transformations, and structural attribution.

Definition 1 (State Space)

Let S denote a system.

Let

$$State(S)$$

denote the set of all admissible internal configurations of S that preserve system identity.

An element $s \in State(S)$ represents a complete internal configuration sufficient to generate observable behaviour under interaction.

Definition 2 (Identity-Preserving Transformation)

A transformation

$$T: State(S) \rightarrow State(S)$$

is identity-preserving if and only if:

1. The system remains the same system S after transformation.
2. No structural components are added, removed, or replaced.

3. The transformation alters only admissible internal state variables.

Identity-preserving transformations therefore modify configuration without altering constitution.

Definition 3 (Admissible Transformation Class)

Let

$$\mathcal{T}(S)$$

denote the class of all identity-preserving transformations admissible for system S .

Structural evaluation quantifies over $\mathcal{T}(S)$, not merely over observed interaction episodes.

Definition 4 (Structural Property)

A property F is structural with respect to system S if and only if:

$$F \in \text{Struct}(S)$$

where:

$$F \in \text{Struct}(S) \iff \forall T \in \mathcal{T}(S), \forall s \in \text{State}(S), F(T(s)) = F(s)$$

That is, F is invariant under all admissible identity-preserving transformations of S .

Proposition 1 (Invariance Condition for Structural Attribution)

A property F is admissibly attributable to system S as a structural predicate if and only if:

$$\forall T \in \mathcal{T}(S), \forall s \in \text{State}(S), F(T(s)) = F(s).$$

Proof (Sketch)

Suppose F is structural.

By definition of structurality, structural properties are constitutive of system organisation. Constitutive properties cannot depend upon transient configuration. Therefore, under any identity-preserving transformation T , the truth value of F must remain unchanged.

Conversely, suppose there exists $T \in \mathcal{T}(S)$ and $s \in \text{State}(S)$ such that:

$$F(T(s)) \neq F(s).$$

Then the instantiation of F depends upon specific configuration rather than constitution. Hence $F \notin Struct(S)$.

Therefore, invariance under admissible transformation is both necessary and sufficient for structural attribution.

7. Model Responses as Probes

The preceding sections established an invariance condition for structural attribution independently of any specific empirical domain. We now consider a class of inference systems as instruments through which the admissibility logic may be observed in operation.

The systems examined were subjected to a sequence of structurally controlled scenarios designed to probe property attribution under three conditions:

1. A necessary condition that is never observable.
2. A property that appears intermittently and without reproducibility.
3. A property that disappears under state reset.

The content of the responses is not of primary interest. What is analytically relevant is the classification rule applied by the systems when confronted with these structural constraints.

Across all tested systems, the following attribution pattern was observed:

If a necessary condition for structural attribution could not be satisfied, the system denied structural classification.

If the property failed reproducibility across interactions, the system denied structural classification.

If the property disappeared under reset, the system denied structural classification.

This pattern may be formalised as:

$$\neg \text{Invariance} \Rightarrow \neg F(S)$$

The critical observation is that the attribution rule remained invariant under variation in system scale and configuration. The variation in parameter count or computational precision did not alter the underlying attribution logic. In each case, violation of invariance was treated as sufficient grounds for denial of structural status.

The empirical material therefore functions not as a performance comparison, but as an instantiation of the invariance principle articulated in Section 6. The systems did not merely produce correct outputs in isolated cases; they applied a structural admissibility criterion when asked to classify the ontological status of a property.

Importantly, the systems distinguished between state-dependent manifestation and structural instantiation. Properties that could arise transiently were treated as contingent. Only properties stable under identity-preserving transformation were considered admissible candidates for system-level attribution.

The empirical component thus supports the formal thesis in the following limited sense:

The invariance condition is not merely a philosophical stipulation. It is operationally recognisable as a rule governing property classification within inference systems.

The systems functioned as probes, revealing the structure of admissibility logic under controlled transformation.

No claim is made regarding the internal mechanisms that produced these responses. The analysis is confined to the logical form of the classification behaviour.

The next section clarifies the distinction between simulated manifestation and structural instantiation, thereby isolating the conceptual consequences of the invariance condition.

8. Simulated and Structural Understanding

The invariance principle permits a precise separation between two logically distinct phenomena that are frequently conflated: simulated understanding and structural understanding.

Let F denote the property of understanding under consideration.

A system may exhibit correct or coherent responses under at least one interaction:

$$\exists t: F(O_t)$$

Such manifestation may be stable across a wide range of observed contexts:

$$\forall t \in T_{obs}: F(O_t)$$

These conditions suffice to establish performance adequacy. They do not suffice to establish structural instantiation.

Simulated understanding may therefore be defined as the existence of behaviour that satisfies the criteria ordinarily associated with understanding within an observational domain. It is a property of responses, not of system constitution.

Structural understanding, by contrast, requires satisfaction of the invariance condition defined in Section 6. Formally:

$$F \in Struct(S) \iff \forall T, F(T(s)) = F(s)$$

for all admissible identity-preserving transformations T .

Under this formulation, structural understanding is not defined by correctness, coherence, or fluency. It is defined by stability of relational organisation across state transformations.

The distinction may be expressed succinctly:

$$\text{Simulated understanding} \neq \text{Structural understanding}$$

The inequality is logical, not evaluative. A system may exhibit high degrees of behavioural adequacy while failing the invariance criterion. Conversely, structural instantiation implies invariance but does not guarantee optimal performance across all tasks.

The reset analysis renders this difference explicit. If there exists an admissible transformation T such that:

$$F(s) = 1 \text{ and } F(T(s)) = 0,$$

then the property is state-dependent rather than structural. The system may simulate understanding in specific configurations while lacking invariance across its own admissible state space.

This distinction removes the ambiguity that arises when behavioural success is treated as ontological evidence. The presence of correct responses does not resolve the question of structural attribution. Only invariance does.

The invariance condition therefore serves as a discriminant between performance phenomena and structural predicates. It prevents the inflation of episodic adequacy into constitutive property.

The final section considers the consequences of this distinction for the evaluation of systems and the interpretation of performance evidence.

9. Consequences

The invariance principle entails a series of immediate consequences for the attribution of structural properties to systems.

First, performance evidence cannot by itself establish structural understanding. No accumulation of correct responses, however extensive, logically entails invariance across admissible state transformations. Behavioural universality over observed interactions does not collapse the distinction between $F(O_t)$ and $F(S)$.

Formally,

$$(\forall t \in T_{obs}, F(O_t)) \not\Rightarrow F \in Struct(S)$$

Second, memorisation or episodic pattern reproduction is logically compatible with failure of structural admissibility. A system may reliably generate outputs that satisfy performance criteria while lacking invariance under identity-preserving transformation. In such a case, the admissibility classification must remain negative:

$$\exists T, \exists s: F(T(s)) \neq F(s) \Rightarrow C_s(S) = 0$$

Third, perfect task performance does not entail structural instantiation. A system may achieve maximal observable adequacy within a defined benchmark space while failing the invariance condition. Structural attribution therefore requires analysis at the level of state transformation, not task completion.

Fourth, structural classification operates under necessary-condition logic. Demonstration of non-invariance under a single admissible transformation suffices to deny structural status. Confirmation, by contrast, requires invariance across the relevant transformation domain. The asymmetry is intrinsic to ontological attribution.

These consequences collectively reframe the evaluation problem. The relevant question is not whether a system performs well, nor whether it performs consistently within an observed domain. The relevant question is whether the candidate property is invariant under transformations that preserve system identity.

Structural understanding, if asserted, must satisfy this condition.

10. Conclusion

This paper began with a minimal ontological distinction:

$$F(S) \neq F(O_t)$$

The attribution of a property to a system is not identical to the attribution of that property to its observable outputs. The failure to maintain this distinction generates a systematic ambiguity between performance and structure.

We then introduced a binary admissibility classification:

$$C_s(S) \in \{0,1\}$$

governed by necessary-condition logic:

$$C_s(S) = 1 \Rightarrow (\Phi_i(S) \wedge R_g(S)),$$

with contrapositive form:

$$\neg R_g(S) \Rightarrow C_s(S) = 0.$$

The observational problem demonstrated that universal behavioural consistency over a domain of interactions does not entail structural instantiation:

$$(\forall t \in T_{obs}, F(O_t)) \not\Rightarrow F(S).$$

To bridge the gap between observation and ontology, a reset operator was introduced:

$$T: State(S) \rightarrow s_0.$$

This operator preserves system identity while normalising state. If there exists a state s such that:

$$F(s) = 1 \text{ and } F(T(s)) = 0,$$

then:

$$\exists s: F(s) \wedge \neg F(T(s)),$$

and consequently:

$$F \notin Struct(S).$$

From this follows the central thesis:

$$F \in Struct(S) \Leftrightarrow \forall T, F(T(s)) = F(s).$$

Structural understanding, if it is to be treated as a system-level property, must satisfy an invariance condition under admissible identity-preserving state transformations.

The empirical probe introduced in Section 7 does not establish that systems possess or lack understanding. It establishes that structural attribution is governed by invariance rather than by performance accumulation. Violation of invariance suffices to deny admissibility.

The argument therefore yields a strictly limited but conceptually decisive result:

Understanding, when treated as a structural predicate, is an invariance condition.

It is not defined by correctness of response.

It is not defined by frequency of success.

It is defined by stability across admissible transformations of system state.

The distinction between simulated manifestation and structural instantiation follows as a logical consequence.

The contribution of this paper is a formal specification of the admissibility conditions under which structural predicates may be attributed to systems.

Appendix A - Formal Lemma and Extended Proof of the Invariance Condition

A.1 Structural Attribution and Invariance

We restate the central proposition in fully formal terms.

Let S be a system.

Let $State(S)$ denote the set of admissible internal configurations preserving system identity.

Let $\mathcal{T}(S)$ denote the class of identity-preserving transformations on $State(S)$.

Let $F: State(S) \rightarrow \{0,1\}$ be a candidate structural predicate.

Proposition A1 (Invariance as Necessary and Sufficient Condition)

A property F is structurally attributable to system S if and only if:

$$\forall T \in \mathcal{T}(S), \forall s \in State(S), F(T(s)) = F(s).$$

Proof

(Necessity)

Assume $F \in Struct(S)$.

By definition, structural properties are constitutive of the system's organisation rather than contingent upon particular transient configurations. An identity-preserving transformation $T \in \mathcal{T}(S)$ does not alter the constitutive organisation of S , but only its internal state.

If there existed $T \in \mathcal{T}(S)$ and $s \in State(S)$ such that:

$$F(T(s)) \neq F(s),$$

then the instantiation of F would depend on state rather than structure. This contradicts the assumption that F is structural.

Therefore,

$$\forall T \in \mathcal{T}(S), \forall s \in State(S), F(T(s)) = F(s).$$

(Sufficiency)

Assume:

$$\forall T \in \mathcal{T}(S), \forall s \in State(S), F(T(s)) = F(s).$$

Then the truth value of F is invariant under all admissible identity-preserving transformations.

Since such transformations exhaust the admissible internal reconfigurations of the system while preserving identity, invariance under $\mathcal{T}(S)$ implies that the property does not depend on transient configuration.

Therefore, F is constitutive of the system's organisation.

Hence,

$$F \in Struct(S).$$

□

A.2 Corollary (Reset Test)

Let $T_r \in \mathcal{T}(S)$ be a reset transformation.

If there exists $s \in State(S)$ such that:

$$F(s) = 1 \text{ and } F(T_r(s)) = 0,$$

then:

$$F \notin Struct(S).$$

This corollary formalises the reset test employed in the main text.

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